

A Concept is More Than a Word: Diversified Unlearning in Text-to-Image Diffusion Models

The Complete Rebuttal

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1 Global Response

We sincerely thank all reviewers for their time, careful reading, and constructive feedback. Below we summarize the key contributions of our work and highlight the additional experiments conducted to address the reviewers’ concerns.

Summary of Contributions. Most existing concept unlearning methods adopt a *keyword-based* formulation: a concept is represented by one or a few textual tokens. However, visual concepts are inherently multi-dimensional and manifest in numerous textual forms. Relying solely on keywords leads to two well-known failure modes: (1) **vulnerability to recovery attacks**, since bypassing a keyword-level filter is straightforward by rephrasing; and (2) **inadvertent damage to related benign concepts**, because a single keyword over-generalizes the unlearning signal beyond the intended target.

Our paper proposes **Diversified Unlearning**, a novel simple-yet-effective unlearning framework that addresses both issues by representing the target concept through a rich set of contextual prompts rather than a fixed keyword. More specifically, the framework introduces a principled mechanism: replacing a single representation of the target concept with a set of rich contextualized representations drawn from natural contexts. Our method diversifies the target concept, broadens the coverage of unlearning, and makes erasure harder to bypass through simple rephrasing or recovery attacks. Moreover, by structuring the erasure loss to align the target with its anchor under varied contexts, our method reduces collateral forgetting through contextually grounded comparisons, helping preserve related but distinct concepts. This can be considered as a contrastive supervision directly within the forgetting objective. Our method is model-agnostic and operates as a **plug-and-play add-on** compatible with both output-based and attention-based unlearning methods, without requiring method-specific tuning.

Reviewer Acknowledgements. The reviewers broadly recognize the paper’s novelty, clarity, and practical value. Reviewer jDEt acknowledges that ‘*Diversified Unlearning is the first distributional framework for concept unlearning in text-to-image diffusion models*’ and finds the experimental setup sound and the results convincing across 10 unlearning methods in five target scenarios. Reviewer 3F2j highlights the framework’s strength as a *lightweight add-on* to both major unlearning pipelines, commends the tailored design for each pipeline’s unique mechanics, and notes the breadth of evaluation across five diverse, high-stakes unlearning settings. Reviewer 3q3q acknowledges that the paper ‘*identifies a real weakness in current concept unlearning pipelines*’ and that the ablation studies on prompt complexity, quantity, and embedding-mixup coefficient provide useful clarity on when diversification helps and when it can start to hurt. Reviewer cxwS praises the clear motivation, strong empirical evaluation, and practical applicability of the modular design for real-world generative model safety. In summary, the reviewers consistently acknowledge the paper’s novelty, strong empirical evaluation, and practical value, while also recognizing Diversified Unlearning as a lightweight and broadly applicable framework.

Key Results Highlighted in Response to Reviewer Concerns. (1) *Generalization to adversarial unlearning baselines (Reviewers cwwS, Q6).* To address concerns about generalization to robustness-oriented methods, we draw attention to results already presented in Table 7 of our submitted paper’s Appendix, where our framework is integrated with **RECE** (attention-based, closed-form) and **Receler** (output-based, adversarial prompt training) on the celebrity concept ‘Henry Cavill’. Di-RECE improves over RECE on both erasure metrics while maintaining preservation, and Di-Receler achieves an 8.53% GPT-score gain in preservation while maintaining erasure strength. Crucially, since Receler already incorporates adversarial prompt augmentation by design, these gains confirm that our improvements stem from the *distributional formulation* rather than prompt diversity alone—directly addressing the concern raised by multiple reviewers.

We believe our responses thoroughly address the reviewers’ remaining concerns and further strengthen the validity and contributions of the proposed Diversified Unlearning framework.

2 Reviewer jDEt

Paper Strengths:

1. As the authors claim, Diversified Unlearning is the first distributional framework for concept unlearning in text-to-image diffusion models.
2. Overall, the paper is clearly written and easy to follow.
3. The experimental setup is sound, including comparisons with state-of-the-art erasure techniques (ESD, UCE, AP, AGE, and ACE) and evaluation across 10 unlearning methods in five target scenarios. The results are convincing.

Response: We thank the reviewer for the time and effort devoted to reviewing our paper. We also appreciate the positive comments about the novelty of our method and the breadth of our experimental evaluation. We would like to address the remaining concerns below:

W1) Although interesting and potentially novel, the proposed approach builds upon and extends existing methods.

Answer:

We agree with the reviewer that our framework is designed to build upon and extend existing unlearning methods rather than replace them. Our goal is to provide a modular mechanism that can be integrated into a diverse set of unlearning pipelines, improving their reliability and stability across different settings.

That said, we would like to clarify that the contribution goes beyond incremental extension. The key novelty lies in reformulating how the target concept is represented in the unlearning objective. Instead of relying on a single fixed prompt or embedding, our method represents the target concept through context-dependent variants, yielding a more diversified unlearning signal that better reflects how concepts appear during pretraining.

More specifically, our **Diversified Prompting / Diversified Embedding Mixup** framework can be viewed as introducing **contrastive structure within the erasure loss**. In the output-based setting, the loss aligns {context + target} with {context + anchor}, which encourages the model to remove the target concept while preserving the surrounding context. In the attention-based setting, directly using contextual prompts leads to severe over-forgetting, so we introduce **Diversified Embedding Mixup** to inject contextual diversity in embedding space while preserving the identity of the target concept.

This differs from conventional data augmentation, which mainly increases input diversity around an otherwise unchanged forgetting objective. In contrast, our framework changes the representation of the target concept inside the objective itself, making erasure broader, more robust to semantic variation, and less harmful to related benign concepts.

Empirically, this formulation improves both output-based and attention-based unlearning methods without method-specific tuning, as demonstrated across multiple baselines in our experiments. We therefore view the contribution not merely as an extension of prior methods, but as a more general framework for making concept-level unlearning more robust and precise.

W2) The method relies on the appropriate selection of a context set for each target concept, which is generated using LLMs.

Answer:

We agree that the selection of context sets is an important component of the framework. To ensure transparency and reproducibility, we provide the context-construction procedure in Appendix A, together with the rule-based

prompt design used to generate progressively richer contexts (see Appendix A.1), including a celebrity concept example in Appendix A.2.1.

More specifically, we leverage LLMs primarily as a scalable tool to accelerate the generation of diverse contextual prompts, which would otherwise require substantial manual effort. We also introduce a verification pipeline to verify that the target concept is correctly reflected in the original model’s output.

For example, prompts such as ‘*A photo of a nudity human holding a golden retriever puppy*’ or ‘*a nudity human wearing a chef’s hat while cooking pasta*’ may fail to produce images containing the target concept (‘nudity’) when evaluated using Stable Diffusion. In practice, these failures likely stem from co-occurrence biases where such objects are predominantly associated with non-nudity images in the pre-training distribution. In such cases, the generated samples do not provide a valid training signal for output-based unlearning methods. The verification step therefore serves to filter invalid prompts and ensure that the resulting context set is both semantically relevant and functionally effective for the unlearning process.

Q1) Please discuss the use of FID in the proposed setting, as it is an evaluation metric that typically requires at least 10k samples to yield reliable results.

Answer: We thank the reviewer for this observation. In our evaluation, preservation is assessed using FID on the COCO-30K benchmark, which contains 30,000 images. This sample size substantially exceeds the commonly recommended minimum of 10k samples for reliable FID estimation, ensuring statistical stability of the reported results.

Importantly, FID is used in the same manner as prior baseline methods to measure preservation performance after removing the target concept ‘*nudity*’. All methods are evaluated using the same COCO-30K set and configuration, ensuring fair and consistent comparison across models.

3 Reviewer 3F2j

Paper Strengths:

1. The framework is designed as a lightweight add-on to both major existing unlearning pipelines (output- and attention-based).
2. The authors avoid the one-size-fits-all pitfalls of many prior methods by tailoring their distributional framework to the unique mechanics of each dominant unlearning pipeline.
3. The evaluation is conducted across five diverse, high-stakes unlearning settings that cover the primary real-world use cases for concept unlearning (copyright protection, harmful content mitigation, celebrity privacy).

Response: We thank the reviewer for the time and effort devoted to reviewing our paper. We also appreciate the positive comments about our framework’s adaptability as a lightweight add-on and its robust performance across a wide range of real-world unlearning use cases. We would like to address the remaining concerns below:

W1) The authors are suggested to conduct computational complexity analysis or run time comparison with the baselines, since the paper frames Diversified Unlearning as a "lightweight add-on" to existing pipelines.

Answer:

Response to Reviewer – Computational Complexity

We thank the reviewer for this suggestion. We provide below a theoretical analysis of the overhead introduced by Diversified Unlearning, supported by empirical runtime comparisons, and clarify why it remains lightweight in both settings.

Diversified Prompting (output-based methods). In the output-based setting, Diversified Unlearning replaces a single keyword prompt c_e with a small set of contextualized prompts $\{c + c_e\}_{c \in \mathcal{C}}$. The per-iteration cost scales by a factor of $|\mathcal{C}|$ over the baseline. In our experiments, $|\mathcal{C}|$ is kept small (e.g., 5–20 prompts), so the absolute overhead is bounded and modest. Crucially, the structure of the loss function — and in particular the diffusion denoising objective — is unchanged; we only diversify the inputs. The dominant cost remains sampling noisy latents z_t across diffusion steps, which is identical to the baseline. Therefore, the asymptotic complexity is $\mathcal{O}(|\mathcal{C}| \cdot T_{\text{baseline}})$, where in practice $|\mathcal{C}|$ is a small constant.

Diversified Embedding Mixup (attention-based methods). In the attention-based setting, our method operates purely in the textual embedding space via a token-level linear interpolation (Equation 7). This mixup is a closed-form operation requiring no additional forward or backward passes through the diffusion model. The

computational overhead added over the baseline is therefore negligible — $\mathcal{O}(d \cdot L)$ per sample where d is embedding dimension and L is sequence length — compared to the dominant cost of computing cross-attention weight updates. The asymptotic complexity is essentially identical to the baseline. To further validate the lightweight nature of our approach, we report wall-clock runtime comparisons between each baseline and its diversified variant in Table 3.

Table 1: Wall-clock runtime comparison between baselines and their Diversified variants.

Setting	Method	Runtime	Overhead
Output-based	AGE	5h 05m 16s	–
	Diversified-AGE	5h 19m 04s	+4.6%
Attention-based	UCE	3s	–
	Diversified-UCE	23s	+20s

In summary, Diversified Unlearning introduces **no new architectural components** and **no additional model parameters**, which is consistent with our characterization as a lightweight add-on. As shown in Table 3, the runtime overhead is minimal: only $\sim 4.6\%$ for output-based methods, and negligible in absolute terms for attention-based methods. We will include this runtime comparison in the revised paper as a dedicated paragraph in Section 4 or the Appendix.

W2) It would be interesting for the authors to further analyze how the framework performs on ambiguous, open-ended concepts that are the highest-stakes for real-world safety.

Answer:

We appreciate this important suggestion. In this work, we evaluate our framework across five distinct concept categories (Table 6 in the Appendix), covering both concrete and abstract / open-ended concepts commonly encountered in real-world safety scenarios:

- **Celebrities:** real human identities (e.g., public figures), which require precise identity-level removal.
- **Objects:** concrete physical entities such as ‘*Garbage Truck*’ or ‘*Cassette Player*’ (Table 9 in the Appendix).
- **Characters:** copyrighted or fictional characters (e.g., ‘*Mario*’), which often involve stylized and context-dependent representations.
- **Nudity:** an inherently **abstract and open-ended safety concept**, where visual boundaries vary across poses, contexts, and artistic styles.
- **Style:** artistic styles such as ‘*Kelly McKernan*’ (Table 13 in the Appendix), which represent high-level visual distributions rather than discrete objects.

Among these categories, **nudity** (Table 3) and **style** (Table 13 in the Appendix) represent the most ambiguous and open-ended concepts, as they are defined by broad semantic patterns rather than fixed visual features. These concepts are particularly challenging for unlearning methods because they require removing distributed semantic attributes while preserving unrelated content.

Diversified prompting and token-level embedding mixup expose the model to a wider range of contextual variations, enabling the erasure mechanism to generalize beyond a single canonical representation of the concept. This property is especially important for open-ended concepts, where the target concept may manifest differently across prompts.

For the **nudity** concept, beyond the I2P dataset results, we further evaluate robustness against recovery attacks using the *Ring-A-Bell* protocol (Table 14 in the Appendix). These experiments demonstrate that our method maintains stable erasure performance even when adversarial prompts attempt to reintroduce the removed concept, suggesting improved robustness in high-stakes safety scenarios.

Overall, these results indicate that the proposed framework is not limited to well-defined object categories, but also generalizes effectively to ambiguous and open-ended concepts that are critical for real-world safety deployment.

W3) Since this is a fast-evolving research direction, several recent works are suggested to be added for further discussion and comparison [1-3].

[1] Unlearning Concepts in Diffusion Model via Concept Domain Correction and Concept Preserving Gradient, AAAI 2025

[2] Meta-Unlearning on Diffusion Models: Preventing Relearning Unlearned Concepts, ICCV 2025

[3] The Illusion of Unlearning: The Unstable Nature of Machine Unlearning in Text-to-Image Diffusion Models, CVPR 2025

Answer:

We thank the reviewer for pointing out these relevant recent works. We are indeed aware of all three papers and agree that they represent important contributions to this rapidly evolving field. Below we clarify the relationship between each suggested work and our proposed method.

[1] **DoCo (AAAI 2025)** propose a concept domain correction framework that (i) aligns output distributions of sensitive and anchor concepts via adversarial training to improve unlearning generalization, and (ii) applies gradient surgery to mitigate conflicts between unlearning and preservation objectives. While DoCo and our work share the motivation of improving generalization beyond the training prompt set, the two approaches are fundamentally different: DoCo operates by modifying the *output distribution* and *gradient directions* during training, whereas our Diversified Unlearning framework targets the *input space* through Diversified Prompting/Diversified Embedding Mixup, and is designed as a *plug-and-play* module compatible with existing unlearning methods. Importantly, our approach can be integrated on top of adversarial-training-based methods (as demonstrated with RECE and Receler in Table 7 of our submitted paper), suggesting complementary rather than competing roles.

[2] **Meta-Unlearning (ICCV 2025)** address a distinct but related problem: preventing *relearning* of unlearned concepts under malicious fine-tuning. Their meta-unlearning framework introduces an additional meta-objective so that, should the model be maliciously fine-tuned on forgotten data, related benign concepts are triggered to self-destruct, hindering concept relearning. This is orthogonal to our work: we focus on improving the *breadth* of concept erasure and preservation quality during the unlearning process by introducing Diversified Prompting/Diversified Embedding Mixup on top of existing baselines, whereas meta-unlearning focuses on *post-release robustness* against adversarial model updates. Notably, their framework is also described as compatible with existing unlearning methods—making it a potentially complementary direction to ours.

[3] **The Illusion of Unlearning (CVPR 2025)** provide an important empirical study demonstrating that unlearned concepts can *revive* upon subsequent fine-tuning of the model, even with general or unrelated prompts. This finding highlights a systemic weakness across existing unlearning techniques, including those we build upon. Rather than contradicting our contributions, this work actually reinforces our motivation: more robust generalization at the prompt level—as provided by our diversified formulation—is a necessary step toward more reliable unlearning. We note that improving resilience against post-release fine-tuning (as studied in [2] and [3]) is an important future direction that goes beyond the scope of our current work.

W4) The paper omits a discussion of the dual-use risk of the proposed framework. The absence of a discussion of the ethical risks and potential guardrails is a notable oversight.

Answer:

We thank the reviewer for raising this important point. We would like to clarify that the primary purpose of our framework is to support responsible deployment of generative models by enabling the removal of sensitive, copyrighted, or harmful content when required. In this sense, ethical considerations are inherent to the motivation for unlearning itself, rather than an external concern introduced by our method.

Our work does not aim to expand model capabilities, but rather to provide a more reliable mechanism for enforcing existing safety and compliance requirements, such as preventing the generation of copyrighted material, sensitive personal data, or inappropriate content. From this perspective, successful unlearning strengthens model safety rather than weakening it.

Nevertheless, we agree that explicitly discussing responsible usage is valuable. We will add a dedicated paragraph in the final version outlining the intended safety-oriented use cases of unlearning to further clarify the ethical context of our work.

4 Reviewer 3q3q - Novelty 2 - Quality 2 - Confidence - 2

Paper Strengths:

1. The paper identifies a real weakness in current concept unlearning pipelines: a visual concept is broader than a single keyword, and keyword-only erasure can be bypassed by rephrasing or can damage nearby concepts in the latent space. This is a meaningful and practically relevant perspective.
2. In the celebrity setting, several diversified variants show large GPT-score reductions for erasure while improving preservation; in the nudity setting, Diversified-ESD and Diversified-AP substantially reduce NudeNet detections while keeping competitive FID/CLIP. These are meaningful improvements. The paper does not stop at headline numbers: it studies prompt complexity, prompt quantity, and the embedding-mixup coefficient α , which helps clarify when diversification helps and when it can start to hurt.

Response: We thank the reviewer for the time and effort devoted to reviewing our paper. We also appreciate the positive comments about our identification of the limitations in keyword-only erasure, the practical relevance of our diversified approach, and the meaningful improvements in both concept erasure and preservation across various settings. We would like to address the remaining concerns below:

W1) The main empirical claim of “consistent” improvement is overstated. The paper repeatedly claims that Diversified Unlearning consistently outperforms keyword-based methods, but the tables show notable exceptions. For example, in the Mario experiment, Diversified-UCF is worse than UCF on erasure GPT-score (26.58 vs. 21.78), and Diversified-ACE is also slightly worse on erasure GPT-score (1.05 vs. 0.28). In the nudity setting, Diversified-ACE is worse than ACE on total NudeNet detections (49 vs. 39) and FID (16.02 vs. 14.63). These counterexamples weaken the central narrative.

Answer:

We appreciate the reviewer’s careful reading and agree that the term ‘consistent’ was overstated in certain contexts. We acknowledge the specific counterexamples raised and will revise the manuscript to reflect a more nuanced claim. That said, we respectfully argue that the overall empirical picture remains strongly positive, for the following reasons.

Improvements are consistent across the majority of methods and settings. Our method achieves clear and reliable gains across a broad range of scenarios: object, style, and celebrity, copyrighted character, nudity concept erasure across ESD, AP, AGE, as well as on preservation metrics nearly universally. For instance, Di-ESD and Di-AP consistently improve over their respective baselines on both erasure and preservation across all concept types (Tables 1–3 of the main paper) and Table 6 in the Appendix.

On the nudity setting specifically. We note that in the adversarial Ring-A-Bell evaluation (Table 7), both Di-ACE and Di-UCF achieve strong erasure performance, with NudeNet detection scores that remain competitive with or better than their baselines under adversarial prompting. The regression in the standard NudeNet evaluation (49 vs. 39) occurs in a regime where ACE already achieves near-complete erasure, leaving minimal headroom for further improvement.

W2) Robustness against recovery attacks is presented as a major contribution, yet the main robustness table only reports AP vs. Diversified-AP, rather than a comprehensive comparison across all baselines. Since robustness is a central claim, the evidence in the main paper feels incomplete.

Answer: We appreciate the suggestion regarding the performance of recovery attacks across all baselines. We would first like to note that the results under Ring-A-Bell attacks and Indirect Recovery (IR) attacks have already been discussed in Appendices B.3 and B.5. To address the reviewer’s concern, we further provide additional experiments under noise-based attacks to better evaluate robustness.

Ring-A-Bell Attacks: With experimental setting provided in Appendix A.1.6, we illustrate the robustness results in Appendix B.5, Table 14 and Figure 6. Specifically, Diversified variants (UCF, ESD, and AP) consistently outperform their original methods under Ring-A-Bell attacks across all K settings, with Diversified-ESD showing the largest ASR reductions and strongest robustness. Diversified-UCF and Diversified-AP also achieve steady improvements, especially at stricter thresholds. Diversified-AGE performs better at higher thresholds despite mixed results at lower ones, while Diversified-ACE demonstrates strong stability, achieving near-zero ASR at K16 and remaining competitive at larger K values.

Indirect Recovery Attacks: Under experimental setup reported in Appendix A.1.3, we present our erasure performance in Appendix B.3 and Table 12, where the Erasure Success Rate under Indirect Recovery Attacks consistently corresponds to the Level 4 metric across all baselines.

Noise-based Attacks: As reported in above table, we observe consistent enhancements across all diversified variants with higher GPT-scores under the ten-celebrities settings.

W3) Context construction uses an LLM plus a verification pipeline based on generated images and a recognition model. This is an important part of the method, but the exact generation rules, filtering thresholds, and sensitivity to the chosen LLM/verifier are not fully analyzed in the main paper. That raises reproducibility concerns and leaves open the possibility that performance depends significantly on prompt-engineering quality.

Answer:

We appreciate the concern about reproducibility and would like to clarify several points that we believe address the raised questions.

The construction pipeline is fully documented and reproducible. The context set construction process is described in detail in Appendix A. Specifically, we provide: (i) a unified prompt-design rule with level-structured generation guidelines (Level-0 through Level-4, Appendix A.2.1), (ii) one illustrative example per setting to guide the LLM, and (iii) the exact verification protocol, using Qwen2.5-VL-72B as the recognition model with a scoring rubric ranging from 0 to 4, accepting only prompts that consistently produce the target concept with high confidence across multiple random seeds. These details are sufficient for independent reproduction. The claim that exact generation

Table 2: Robustness evaluation of unlearning methods under Noise-based attacks (Noise ATK). We report GPT-score on ten-celebrity erasure.

Method	Noise ATK (GPT↓)
AP	39.63
Diversified-AP	14.94
ESD	28.19
Diversified-ESD	11.75
AGE	33.00
Diversified-AGE	29.81
UCE	2.00
Diversified-UCE	1.13

rules are absent does not align with the content of Appendix A.

The method is robust to prompt diversity, not dependent on prompt perfection. core concern raised is that performance may hinge on prompt-engineering quality. We argue the opposite is true by design. As shown in the ablation study (Appendix A.2.3), our method yields consistent improvements across varying numbers of prompts (5, 10, 20, 50) and across prompt complexity levels (Level-1 through Level-4). This indicates that the framework is stable across a wide range of prompt configurations, rather than sensitive to any particular prompt choice. Furthermore, we explicitly and intentionally adopt a generic, concept-agnostic prompt construction strategy to ensure scalability — the goal is breadth of coverage, not precision of any individual prompt.

Sensitivity to the LLM or verifier is not a fundamental concern, given the method’s distributional objective. This is perhaps the most important point. As established in Section 4, visual concepts are inherently multi-dimensional — a concept cannot be fully captured by any single textual expression. This is precisely why keyword-based unlearning is brittle: it relies on a narrow point estimate of the concept’s semantic space. Diversified Unlearning is designed to shift from this point estimate to a distributional representation, sampling diverse contexts that collectively approximate the concept’s true semantic breadth. Under this distributional framing, the goal of context set construction is not to find the perfect set of prompts, but to obtain a sufficiently diverse sample from the concept’s contextual distribution. Consequently, even if a different LLM or verifier produces a somewhat different context set, the unlearning signal remains effective as long as the resulting prompts cover varied contextual manifestations of the concept. The method’s robustness is therefore inherent to its distributional design, not contingent on any specific LLM or verifier choice.

W4) The high-level insight is interesting, but the actual mechanism is largely a combination of contextual prompt augmentation and embedding interpolation on top of existing methods. The paper’s contribution is more of a strong empirical design pattern than a fundamentally new unlearning principle.

Answer:

We agree that our method is designed as a modular add-on that can be integrated into existing unlearning frameworks. However, we respectfully clarify that the contribution is not merely a heuristic combination of prompt augmentation and interpolation, but a principled mechanism that reshapes the optimization dynamics of the erasure objective.

As discussed in Section 3 (Background and Preliminaries), a typical unlearning objective consists of two components: the *erasure loss* (L_1) and the *preservation loss* (L_2), each serving distinct roles. Conventional approaches primarily rely on L_2 to maintain model capability, while L_1 focuses solely on removing the target concept. In contrast, our **Diversified Prompting / Diversified Embedding Mixup** formulation can be viewed as introducing **contrastive structure within the erasure loss itself**, thereby improving preservation implicitly through the forgetting process.

To illustrate the limitation of keyword-based unlearning, consider the task of forgetting the target concept ‘*Henry Cavill*’. In standard keyword-based settings, the input prompt ‘*Henry Cavill*’ is mapped to a neutral anchor concept c_a (e.g., ‘a person’ or an empty neutral string). This setup introduces two practical limitations.

- First, in output-based methods such as ESD, AP, and AGE, the model observes generated images that may contain peripheral details unrelated to the target concept. During standard unlearning, the optimization process may inadvertently map both the target concept and these contextual details toward the neutral anchor concept c_a , thereby degrading preservation performance.

- Second, the prompt ‘*Henry Cavill*’ is encoded into a single fixed embedding that remains constant across unlearning iterations or closed-form updates. This contrasts with the original diffusion model training process, where the same concept appears in diverse contexts, producing context-dependent embedding variations. The mismatch between static unlearning inputs and context-rich pretraining data can limit the effectiveness and robustness of erasure.

The key insight of the proposed framework is therefore to introduce **controlled contrast within the erasure loss**.

- By augmenting prompts with contextual information—forming pairs of {context + target concept} mapped to {context + neutral anchor concept} (L_1 in Eq. 5)—the model is explicitly encouraged to remove only the target concept while preserving the surrounding context. This structured contrast sharpens the optimization signal in L_1 , enabling more precise concept removal and improving preservation without relying solely on an additional preservation loss.
- Furthermore, prompts of the form {context + target concept} produce **context-sensitive embeddings** through interactions within the text encoder. This diversification increases the representation coverage of the target concept in embedding space, leading to more stable and generalizable unlearning behavior across semantic variations.

In this sense, our contribution can be interpreted not simply as an empirical design pattern, but as a mechanism that modifies how the erasure objective is structured—introducing contrastive supervision directly into the forgetting process. This formulation is model-agnostic and consistently improves both output-based and attention-based unlearning methods without method-specific tuning, as demonstrated across multiple baselines in our experiments.

5 Reviewer cxwS - Novelty 2 - Quality 2 - Confidence - 3

Paper Strengths:

1. Clear motivation. The paper clearly identifies a key limitation of existing concept unlearning methods: the overly simplistic keyword-based representation of visual concepts.
2. Strong empirical evaluation. The paper provides extensive experimental results across multiple concept erasure settings, including celebrities, copyrighted characters, and explicit content.
3. Practical applicability. The proposed framework is designed as an add-on module that can be integrated with existing unlearning methods. This modular design makes the approach easy to adopt without requiring significant architectural modifications, increasing its practical usefulness for real-world generative model safety.

Response: We thank the reviewer for the time and effort devoted to reviewing our paper. We also appreciate the positive comments about our clear motivation to address the limitations of keyword-based representations, the extensiveness of our empirical evaluation, and the practical modularity of our proposed framework. We would like to address the remaining concerns below:

W1) Limited justification of the “distributional” formulation. The paper frames the proposed approach as a distributional formulation of concept unlearning by representing a concept with a set of contextual prompts. However, the paper does not clearly justify why the constructed context set meaningfully approximates the underlying semantic distribution of the concept. Since the prompts are generated heuristically and filtered through a verification pipeline, it is unclear whether this process actually captures the concept distribution or simply introduces prompt diversity. As a result, the improvement may be attributed to prompt augmentation rather than a principled distributional formulation.

Q1) Justification of the “distributional” formulation. The paper describes Diversified Unlearning as a distributional formulation of concept erasure by introducing a set of contextual prompts. However, it is unclear why this context set can be considered a meaningful approximation of the underlying concept distribution. In practice, the context prompts appear to be generated heuristically (e.g., using an LLM), and it is not clear whether they truly capture the semantic variability of the concept. Furthermore, it would be helpful to clarify whether the context set is constructed independently from the target concept and the anchor concept. If the contexts are strongly coupled with the target prompts, it is unclear in what sense the method represents a distribution rather than simply

augmenting the training examples. Relatedly, it would be helpful to clarify how this approach differs from standard prompt augmentation or adversarial prompt training. The observed improvements may partly come from increased prompt diversity or additional training signals rather than from a principled distributional formulation.

Answer:

Justification of the Distributional Formulation (W1, Q1). We thank the reviewer for this question and take the opportunity to clarify the conceptual foundation of our approach.

Why the context set approximates the concept distribution. We agree that the term ‘distributional’ warrants more careful justification. Our key observation is that during pretraining, a concept such as ‘*Henry Cavill*’ is never encountered in isolation—it always appears embedded in diverse linguistic contexts (e.g., ‘a photo of Henry Cavill as Superman,’ ‘Henry Cavill at a red carpet event,’ etc.). As a result, the text encoder learns *context-sensitive representations* of the concept: the same token sequence produces different embedding vectors depending on its surrounding context. Standard keyword-based unlearning collapses this rich distribution into a *single fixed embedding*, creating a mismatch between the static unlearning signal and the context-rich pretraining distribution. Our contextual prompt set—generated via LLM and verified through a filtering pipeline—is therefore not intended to be an exact distributional model, but rather a practical approximation of this pretraining variability that **increases embedding coverage** of the target concept in the encoder’s representation space.

How this differs from prompt augmentation or adversarial prompt training. This distinction is crucial. Standard prompt augmentation or adversarial prompt training (e.g., Receler, AdvUnlearn) introduces diverse prompts primarily to *stress-test* the erasure objective—ensuring the model cannot regenerate the concept under paraphrased inputs. Our approach is fundamentally different in mechanism: by forming pairs of $\{\text{context} + \text{target concept}\} \rightarrow \{\text{context} + \text{anchor concept}\}$, we introduce **contrastive structure within the erasure loss itself**. This encourages the model to remove only the target concept while explicitly *preserving the surrounding context*, which directly improves preservation without relying solely on a separate preservation loss L_2 . In other words, the improvement does not merely come from seeing more prompts, but from how those prompts reshape the optimization geometry of L_1 . As supporting evidence, our method consistently improves preservation even when integrated on top of methods that already incorporate adversarial prompt training (Di-Receler, Table 7), demonstrating gains beyond what prompt diversity alone can explain.

On the independence of the context set from the target and anchor concepts. The context prompts are generated conditioned on the *category* of the target concept (e.g., “celebrity,” “artistic style”), not on the specific target or anchor identity. This ensures that the context set captures general semantic variability of the concept type rather than being tightly coupled to any particular target–anchor pair. The Diversified Embedding Mixup further reinforces this independence: the mixed embedding $y = \alpha x + (1 - \alpha)z$ is constructed in embedding space, where z can be either a context-driven token embedding, ensuring that the perturbation is not simply a restatement of the target prompt.

W2) Dependence on heuristic context generation. The context set is generated using an LLM and a verification pipeline. This introduces a heuristic component that may influence the effectiveness of the method. The paper does not analyze how sensitive the method is to the quality or diversity of these generated prompts, which raises concerns about robustness and reproducibility.

Answer: We acknowledge that the context set is generated using an LLM together with a verification pipeline, which introduces a heuristic component. However, this design is motivated by a practical limitation observed in output-based unlearning methods. For example, prompts such as ‘*A photo of a nudity human holding a golden retriever puppy*’ or ‘*a nudity human wearing a chef’s hat while cooking pasta*’ may fail to produce images containing the target concept (‘nudity’) when evaluated using Stable Diffusion. In such cases, the generated samples do not meaningfully activate the target concept, reducing the effectiveness of erasure for output-based methods that rely on model responses to these prompts. The verification pipeline therefore serves to ensure that each prompt is semantically valid and capable of eliciting the target concept, rather than introducing arbitrary variation.

We further analyze the impact of prompt diversity and complexity in Figure 4. As shown in the figure, diversified prompting consistently outperforms the keyword-only baseline even at Level-1, demonstrating the immediate benefit of introducing contextual variation. Increasing prompt complexity improves performance up to a moderate level (Level-3), which yields the best erasure–preservation trade-off. In contrast, excessive complexity (Level-4) leads to degraded erasure performance, suggesting that the method is not overly sensitive to prompt diversity but instead exhibits a predictable performance trend with respect to prompt complexity.

Q2) Scalability and cost of context generation. The proposed method requires constructing a context set for each target concept. Does this process need to be repeated whenever a new concept is selected for unlearning? If so, what is the computational overhead and wall-clock time required to construct these context sets?

Answer: We clarify that context generation is **not** a per-concept bottleneck. We maintain a set of diversified concept templates categorized by type. For any new entity within a known category (e.g., a new celebrity), we simply perform a string-matching replacement.

W3) Limited analysis of computational overhead. The proposed method requires constructing context sets and sampling contextual prompts during training. However, the paper does not provide detailed comparisons of training complexity or wall-clock time relative to baseline methods. This makes it difficult to assess the computational cost of the proposed framework in practice.

Q3) Training complexity and computational overhead. The loss function involves sampling contextual prompts from the context set during training. Could the authors provide more details on the training complexity compared to baseline methods, including wall-clock training time and the number of context samples used per target concept?

Answer:

We provide below a theoretical analysis of the overhead introduced by Diversified Unlearning, supported by empirical runtime comparisons, and clarify why it remains lightweight in both settings.

Diversified Prompting (output-based methods). In the output-based setting, Diversified Unlearning replaces a single keyword prompt c_e with a small set of contextualized prompts $\{c + c_e\}_{c \in \mathcal{C}}$. The per-iteration cost scales by a factor of $|\mathcal{C}|$ over the baseline. In our experiments, $|\mathcal{C}|$ is kept small (e.g., 5–20 prompts), so the absolute overhead is bounded and modest. Crucially, the structure of the loss function — and in particular the diffusion denoising objective — is unchanged; we only diversify the inputs. The dominant cost remains sampling noisy latents z_t across diffusion steps, which is identical to the baseline. Therefore, the asymptotic complexity is $\mathcal{O}(|\mathcal{C}| \cdot T_{\text{baseline}})$, where in practice $|\mathcal{C}|$ is a small constant.

Diversified Embedding Mixup (attention-based methods). In the attention-based setting, our method operates purely in the textual embedding space via a token-level linear interpolation (Equation 7). This mixup is a closed-form operation requiring no additional forward or backward passes through the diffusion model. The computational overhead added over the baseline is therefore negligible — $\mathcal{O}(d \cdot L)$ per sample where d is embedding dimension and L is sequence length — compared to the dominant cost of computing cross-attention weight updates. The asymptotic complexity is essentially identical to the baseline. To further validate the lightweight nature of our approach, we report wall-clock runtime comparisons between each baseline and its diversified variant in Table 3.

Table 3: Wall-clock runtime comparison between baselines and their Diversified variants.

Setting	Method	Runtime	Overhead
Output-based	AGE	5h 05m 16s	–
	Diversified-AGE	5h 19m 04s	+4.6%
Attention-based	UCE	3s	–
	Diversified-UCE	23s	+20s

In summary, Diversified Unlearning introduces **no new architectural components** and **no additional model parameters**, which is consistent with our characterization as a lightweight add-on. As shown in Table 3, the runtime overhead is minimal: only $\sim 4.6\%$ for output-based methods, and negligible in absolute terms for attention-based methods. We will include this runtime comparison in the revised paper as a dedicated paragraph in Section 4 or the Appendix.

Q4) Design choice of token-level embedding mixup. The proposed embedding mixup is applied at the token level rather than the prompt level. Could the authors provide further intuition or analysis explaining why token-level interpolation performs better in this setting?

Answer:

Our design choice is motivated by the specific mechanical properties of **closed-form attention-based unlearning**, a category that includes both **UCE** and its robust variant, **RECE**. While RECE incorporates *adversarial training* to identify challenging text embeddings, it shares a common foundation with UCE in its reliance on a **closed-form weight update** within the cross-attention layers. Specifically, these methods execute concept erasure by mapping the text encoder’s representation of a target-conditioned prompt toward a neutral anchor concept c_a , as formulated in Eq. (4).

When a prompt contains multiple objects or contextual elements, this mapping does not act exclusively on the target concept; instead, surrounding contextual tokens may also be shifted toward the anchor concept. Such unintended modification of non-target semantics can degrade preservation performance. For this reason, prior UCE-based approaches typically rely on simple input prompts that isolate the target concept.

Table 4: Quantitative results of erasing ‘Henry Cavill’ using baselines compared with their diversified counterparts (denoted as Di^* , e.g., Di-ESD for Diversified-ESD) across prompt complexity levels. Erasure and preservation are evaluated using GPT-score, CLIP and LPIPS.

Method	Erasure		Preservation			
	CLIP-i↓	GPT↓	LPIPS↓	CLIP-i↑	CLIP-t↑	GPT↑
UCE	62.36	10.25	0.47	86.27	29.99	88.97
Di-UCE	61.23	6.95	0.48	86.11	29.95	90.95
RECE	59.42	8.05	0.79	77.34	29.32	77.42
Di-RECE	58.57	7.78	0.79	77.95	29.64	77.87
Receler	51.61	0.00	0.57	73.93	25.27	47.07
Di-Receler	52.90	0.00	0.52	77.65	26.17	55.60

In this setting, applying diversified prompting at the prompt level—where each prompt introduces different contexts or objects—can amplify this limitation, as more contextual tokens become susceptible to unintended shifts during the closed-form update. Token-level embedding mixup mitigates this issue by restricting interpolation to the representation of the target concept itself, while leaving the surrounding contextual structure relatively stable.

Q5) Sensitivity to the interpolation coefficient. Given that α is extremely close to 1, the mixed embedding remains almost identical to the original concept embedding. It would be helpful to clarify whether the performance gain mainly comes from contextual diversity or simply from regularization effects.

Answer: We thank the reviewer for raising this question. We would like to first point out that this is already discussed in Appendix B.6 (Ablation Study) of our submitted paper. Specifically, the default large value of α may initially seem surprising. While the mixed embeddings may still appear visually similar to the original concept from a human perspective, the behavior is different from the perspective of the text encoder. In practice, this large α provides a favorable trade-off: it preserves sufficient information for the encoder to reliably identify the target concept while still introducing enough diversity to benefit the unlearning process.

Mathematically, consider the mixed embedding

$$y = \alpha x + (1 - \alpha)z, \quad x, y \in \mathbb{R}^d,$$

where z is either the Diversified Token Embedding or the Noise-based Embedding. The signal-to-noise (SNR) ratio scales proportionally to

$$\frac{\alpha^2 \|x\|^2}{(1 - \alpha)^2 d},$$

and the cosine similarity between x and y approximates:

$$\frac{\alpha \|x\|}{\sqrt{\alpha^2 \|x\|^2 + (1 - \alpha)^2 d}}.$$

These expressions illustrate that in very high-dimensional spaces (large d), keeping $(1 - \alpha)$ small is crucial to maintaining both a high SNR and strong alignment between x and y . This explains why a numerically large α remains appropriate despite appearances.

Q6) Additional comparisons. The experimental evaluation appears to include only one attention-based baseline. It would strengthen the empirical validation if the authors could include additional attention-based methods for comparison. Additionally, comparisons with prompt augmentation or adversarial prompt training would help clarify whether the observed improvements stem from the proposed distributional formulation or simply from increased prompt diversity.

Answer:

In Appendix B, Table 7, we provided experiments with two SOTA adversarial training based unlearning baselines, **RECE** and **Receler**. It can be seen that, our **Di-RECE** improves over RECE on both erasure metrics (CLIP-i: 59.42 $\rightarrow$ 58.57; GPT-score: 8.05 $\rightarrow$ 7.78) while maintaining comparable preservation performance, demonstrating that our approach generalizes effectively to attention-based methods. For **Di-Receler**, integrating our Diversified Unlearning module preserves the original model’s erasure capability while markedly enhancing preservation, yielding an 8.53% improvement in GPT-score (47.07 $\rightarrow$ 55.60). Notably, Receler already incorporates adversarial prompt

augmentation by design; nonetheless, our diversified formulation provides consistent and complementary gains on top of it. This result confirms that the observed improvements are not simply a byproduct of increased prompt diversity, but arise from our *distributional formulation* that explicitly encourages the model to maintain its generative capability across a broader semantic space.